

* Type IIB: Two of the same chirality vec. multiplets

• Cannot be obtained from dim reduction.

$$G_{IIB} = ([1000]_8 + [0010]_8)^2, \quad N = (2,0) \text{ susy}$$

32 - supercharges, chiral (by construction)

$$\begin{aligned} G_{IIB} &= [1000]_8 [1000]_8 + 2 [1000]_8 [0010]_8 + [0010]_8 [0010]_8 \\ &= [2000]_8 + [0100]_8 + [0000]_8 + 2 [1010]_8 + 2 [0001]_8 \\ &\quad + [0010]_8 [0010]_8 \end{aligned}$$

* Calculate: Use triality on

$$\begin{aligned} [1000]_8 [1000]_8 &= [2000]_8 + [0100]_8 + [0000]_8 \\ \rightarrow [0010]_8 [0010]_8 &= [0020]_8 + [0100]_8 + [0000]_8' \end{aligned}$$

$\therefore$

$$\begin{aligned} G_{IIB} &= [2000]_8 + [0100]_8 + [0000]_8 + 2 [1010]_8 + 2 [0001]_8 \\ &\quad + [0020]_8 + [0100]_8 + [0000]_8 \end{aligned}$$

$[0020]$: Self-dual 4-form

$[0002]$: Anti self dual 4-form

* Self-dual 5-form $[0020]_8$ makes it difficult to construct a covariant action

Green-Schwarz string $\rightarrow$ type IIB:

* 5-form gauge field does not contain an action.

$\rightarrow$ Use variant of Noether method

$\rightarrow$ Two scalars: $\in SU(1,1)/U(1)$

• Susy closes on F.C.M only

$$\begin{aligned} Q_\alpha^i, \quad i=1,2 \quad \alpha=1,\dots,32 \quad & \left| \quad (Q_\alpha^i)^* = Q_\alpha^i \right. \\ & \left. \gamma_{\alpha\beta} Q^\beta = Q_\alpha \right. \end{aligned} \quad \epsilon_{ij}$$

$$D = 10: \quad R = 11:$$

$$\{Q_\alpha^i, Q_\beta^j\} = (P_\alpha \gamma_{mC}^{-1})_{\alpha\beta} \delta^{ij} P^m + (P_\alpha \gamma_{mC}^{-1})_{\alpha\beta} Z_m^{ij} + (P_\alpha \gamma_{m_1 \dots m_5 C}^{-1})_{\alpha\beta} Z^{m_1 \dots m_5 ij} \\ + (P_\alpha \gamma_{m_1 \dots m_5 C}^{-1})_{\alpha\beta} Z^{m_1 \dots m_5 ij}$$

$$P_\alpha = \frac{1}{2} (\gamma_{11} - \gamma_{11}), \quad Z_m^{ij} \text{ is symm and traceless}$$

$Z^{m_1 \dots m_5}$: Dual / Anti-self dual.

* Includes an $SO(2) \cong U(1)$ symmetry which rotates the (ij) indices.

(R-Symmetry)

Complex Weyl: $Q_\alpha = Q_\alpha^1 + iQ_\alpha^2$
 $\bar{Q}_\alpha = Q_\alpha^1 - iQ_\alpha^2 = (Q_\alpha)^*$

* $U(1)$ generator: R
 $U(1) \ni g = e^{i\epsilon R}$, with $R^\dagger = -R$

$$[Q_\alpha^i, R] = i\gamma_{11} Q_\alpha^i = iQ_\alpha^i$$

$$R^\dagger Q_\alpha^{\dagger i} - Q_\alpha^{\dagger i} R^\dagger = -iQ_\alpha^{\dagger i}$$

$$= -R \bar{Q}_\alpha^i + \bar{Q}_\alpha^i R = -i\bar{Q}_\alpha^i$$

$\therefore$ $[\bar{Q}_\alpha^i, R] = -i\bar{Q}_\alpha^i$

* $e_m^\mu, A_{\mu\nu}, P, B_{\mu\nu\rho\kappa}, \psi_\alpha^m, \lambda_\alpha$ | ψ_μ, λ_α : $\mathbb{C}$ -valued spinors
 $\begin{pmatrix} A_{\mu\nu} \\ \psi \end{pmatrix}$: Both $\mathbb{Q}$ -valued

$$[2000]_8, [6100]_8, [0000]_8, [0020]_8, [1001]_8$$

$$e_\alpha^m$$

$$A_{\mu\nu}$$

$$\psi$$

$$B_{\mu\nu\rho\kappa}$$

$$\psi_\alpha^m$$

$$\underline{35}$$

$$2 \cdot \underline{28}$$

$$2 \cdot \underline{1}$$

$$35$$

$$\underline{112}$$

$$\frac{D(D-3)}{2}$$

$$\begin{pmatrix} D-2 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} D-2 \\ 4 \end{pmatrix}$$

$$2 \cdot 7 \cdot 2^3$$

$$\underline{2}$$

$$[0001]_8$$

$$\lambda_\alpha$$

$$\underline{2 \cdot 8 = 16}$$

$$G_{m_1 \dots m_5} \equiv 5 \partial_{[m_1} B_{m_2 \dots m_5]}$$

-D Self-dual: * $G_{m_1 \dots m_5} = \frac{1}{5!} G_{m_1 \dots m_5 p} \eta^{m_6 \dots m_{10}} = G_{m_1 \dots m_5}$

* Note $\binom{8}{4} = \frac{8!}{4!4!} = \frac{30720}{4824} = 70/2 = 35$

R-symmetry weights:

$\delta_R E = i p_R E$ 1 p:inf. transf.

E	R	
e_m^μ	0	• Real fields have <u>zero</u> R-symm.
$A_{\mu\nu}$	2	• $[R, Q_\alpha^i] = i (r_\alpha^i) Q_\alpha^i$
ψ	4	$r_{Q_i} =$
B_s^{grav}	0	
ψ_μ^α	2	} opposite } ψ_μ is the Gauge field of R-charge } SUSY.
λ_α	3	
Q_α^i	-1	

* 4-form: $\int d^{10} G_{\mu_1 \dots \mu_5} G^{\mu_1 \dots \mu_5} = \int G \wedge G = 0$

$\Rightarrow$ Thus cannot write the action

Type IIB string theory:

NS $\oplus$ NS	R $\oplus$ R	$A_{\mu\nu} \equiv A_{\mu\nu}^2 + i A_{\mu\nu}^1$
$e_m^\mu, A_{\mu\nu}^2, \phi$	$A_{\mu\nu}^1, \chi, B_{\mu\nu\rho\tau}$	$\psi \equiv \phi + i\chi$
$\uparrow \quad \uparrow$	$\uparrow \quad \uparrow$	
R	R	

* ψ : Belongs to a non-linear realisation of $SU(1,1)$ w/ local $U(1)$.

• In the absence of all other type IIB fields F.O.M. ψ reduce to:

$S_\psi^{\text{free}} = -\frac{1}{2} \int d^2x \frac{\partial_\mu \psi \partial_\mu \psi^*}{(1-|\psi|^2)^2}$ $\leftarrow$ σ -model $SL(2, \mathbb{R})$?

• Only occur from $\mathcal{P}_\mu$?

* $\Lambda : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ = 32. Λ = ...

$\frac{1}{mN} : F_{1\dots m3} = \partial(m_1 A_{m2} m_2)$, $SU(1,1)$ must be matter
 reps and can only transform under rigid $SU(1,1)$
 and inert under $U(1)$

* $g_0 \in SU(1,1)$:

$$\begin{pmatrix} \bar{F}_{mnp} \\ F_{mnp} \end{pmatrix} \rightarrow D(g_0^{-1}) \begin{pmatrix} \bar{F}_{mnp} \\ F_{mnp} \end{pmatrix} \quad \Bigg| \quad D(g_0^{-1}) = g_0$$

$$g_0 = \begin{pmatrix} u & v \\ v^* & u^* \end{pmatrix} \quad \Bigg| \quad u u^* - v v^* = 1$$

* Define: $\begin{pmatrix} \bar{F}_{mnp} \\ F_{mnp} \end{pmatrix} \rightarrow D((g(\varphi))^{-1}) \begin{pmatrix} \bar{F}_{mnp} \\ F_{mnp} \end{pmatrix}$

$$\begin{pmatrix} \bar{F}_{mnp} \\ F_{mnp} \end{pmatrix} \xrightarrow{D(h)} \begin{pmatrix} \bar{F}_{mnp} \\ F_{mnp} \end{pmatrix} \xrightarrow{\quad} \left\{ \begin{array}{l} \bar{F}_{mnp} \rightarrow e^{-2i\alpha} \bar{F}_{mnp} \\ F_{mnp} \rightarrow e^{2i\alpha} F_{mnp} \end{array} \right\}$$

→ From we'll come back to this...

E.O.M + $\mathcal{L}_{\text{susy}}$ given in texts